

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-608

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	608	3FA	350A-WW-SLD-ST1-N-8577-01	1 OF 2	1	CJ-M15W14-PR-A20-000115	2A	
2	608	3FA	350A-WW-SLD-ST1-N-8577-02	2 OF 2	0	CJ-M15W14-PR-A20-000115	2A	
3	608	3FA	350A-WW-SLD-ST1-N-8578-01	1 OF 1	1	CJ-M15W14-PR-A20-000115	2A	
4	608	4FA	350A-WW-SLD-ST1-N-8573-01	1 OF 1	0	CJ-M15W14-PR-A20-000114	1A	
5	608	4FA	350A-WW-SLD-ST1-N-8577-01	1 OF 1	0	CJ-M15W14-PR-A20-000115	2A	



ITEM NO.	W14-AD-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK AIR DIFFUSER
Q'TY	110
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-T-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	R.C + FRP

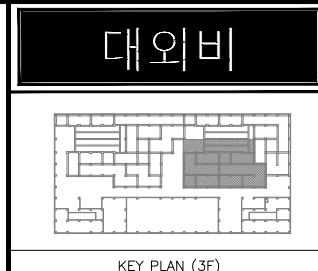
ITEM NO.	W14-T-4230 B
SERVICE	CIF MBR TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

ITEM NO.	W14-T-4232 B
SERVICE	CIF MBR CLEANING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	C.S + FRP

ITEM NO.	W14-AD-4230 B
SERVICE	CIF MBR AIR DIFFUSER
Q'TY	ea
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-EJ-4231 B
SERVICE	CIF MBR EJECTOR UNIT
Q'TY	ea
TYPE	BY VENDOR
CAPA.	N/s.ea
MAT'L	BY VENDOR

ITEM NO.	W14-MBR-4230 B
SERVICE	CIF MBR MEMBRANE
CAPA.	m3/d
	15,000



- NOTE
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치함
 2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성함
 3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
 4. 운전 시 CLOSE 되는 배관에는 LOTO 설치
 5. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
 6. 4부 FRP LINING TANK SLEEVE SUS304 + FRP LINING 적용
 7. PACKAGE VENDOR 선정 후 4부 SYSTEM (공회 EQUIPMENT SPEC., LINE SIZE, MBR Nos., INSTRUMENT CONTROL SCHEME 등) 확정 예정
 8. 안전포기
 9. 차단포기
 10. 용압계 설치
DESIGN RANGE : -1~1 BAR

INTERLOCK NO. 1-4231 B (W14-P-4231 B1/B2 AUTO STOP)
RENOTE STATUS에 W14-T-4229 B LEVEL (LI-W2321) LOW LOW STOP.

12. OPERATION TIME CHART

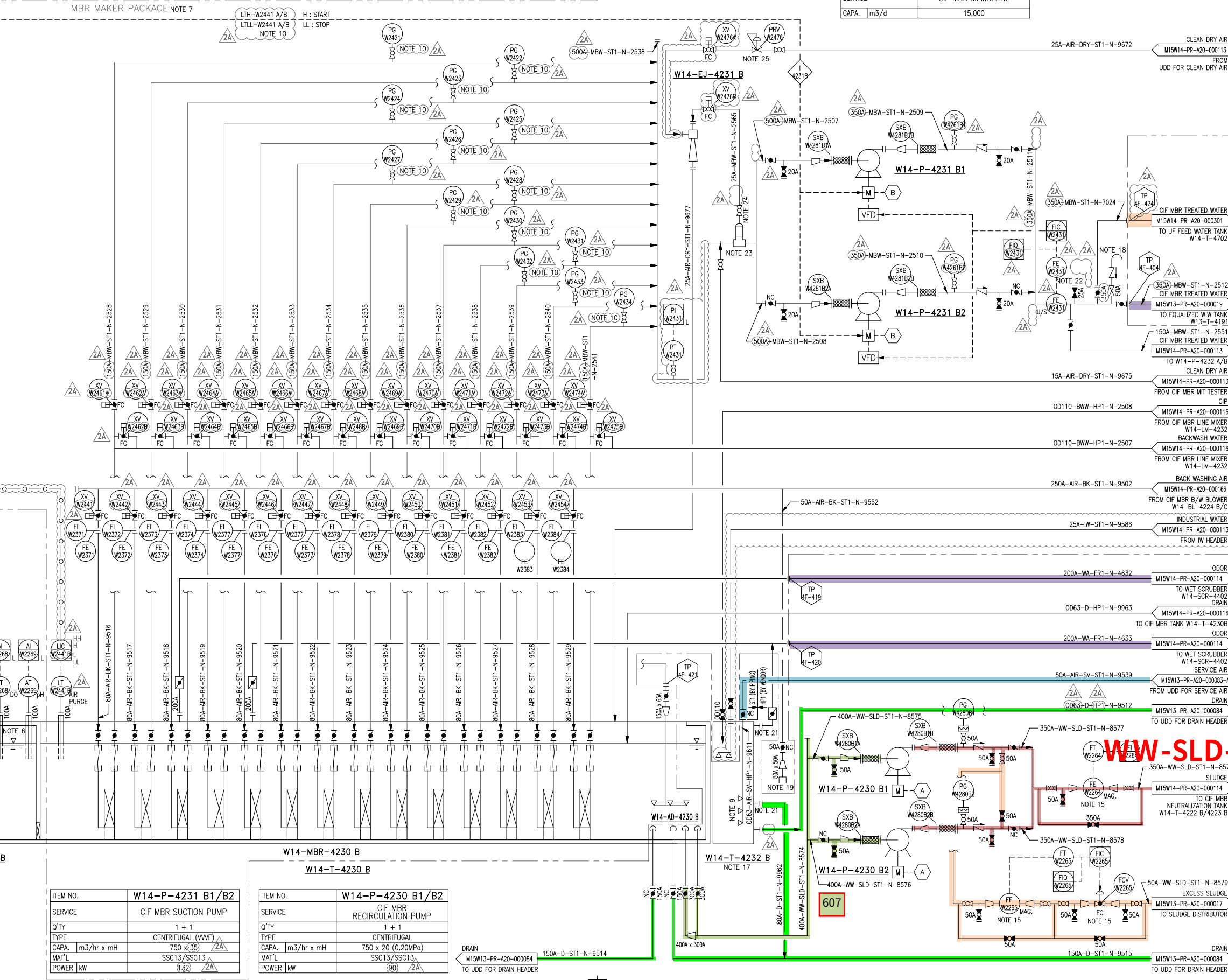
PROCESS	SERVICE	B/W READY	B/W	RESTART
TIME(sec)	24hr	2	30	30
XV-2331A	OPEN	-	-	-
XV-2331B	-	-	OPEN	-
XV-2321	-	OPEN	-	OPEN
B/W PUMP	-	-	RUN	-

11. 4부 ON-OFF VALVE 등도 동일하게 적용한다.
13. CIF MBR TANK CRANE를 PACKAGE VENDOR에게 공급한다.
14. DELETED.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정
16. MANHOLE 2개소 설치하여야 한다.
17. SUPPLIED BY MBR PACKAGE VENDOR.
18. HIGH POINT에 SIPHON BREAKER 설치함.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
19. 개개의 세정용 CASSETTE CONNECTION 용도.
MBR CLEANING TANK 인근에 설치함
20. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치함 (TAG NO. 아래 참조)

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

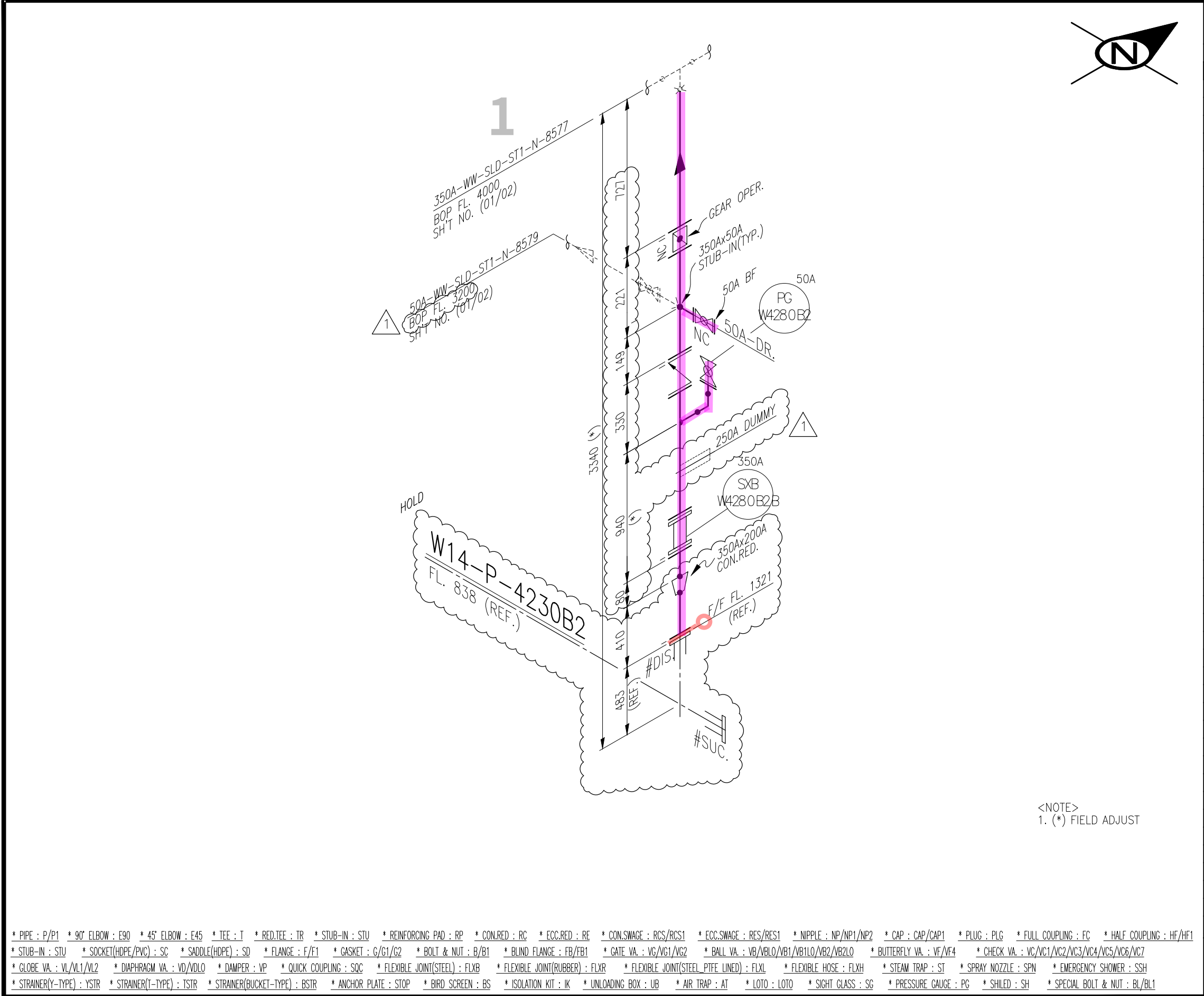
수정	REV.	실명	DESCRIPTION	날짜	DATE
553	도	이인학			
608	검토	한진희			
	확인	박동영			
	승인	사			

도면명	PIPING & INSTRUMENT DIAGRAM
DWG. TITLE	CIF MBR DISCHARGE & CIRCULATION PUMP B (2/2)
SCALE	NONE
날짜	2025. 08. 14
DATE	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000115
SHEET NO.	2A



ITEM NO.	W14-P-4231 B1/B2
SERVICE	CIF MBR SUCTION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(132)

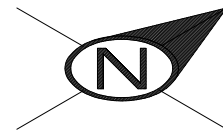
ITEM NO.	W14-P-4230 B1/B2
SERVICE	CIF MBR RECIRCULATION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(90)



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED. : RC * ECC.RED. : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (℃)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.1	60	UNIT	3FA								M15X Ph4 Project	
			OPERATING	2.0	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-030001-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	WW-SLD	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000115								350A-WW-SLD-ST1-N-8578	01 / 01	1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



W14-T-4223 B

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	3.1	60	UNIT 4FA						
			OPERATING	2.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040006-B						
						REF. P & ID CJ-M15W14-PR-A20-000114		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

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CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW-SLD		22330D	
LINE NO.	SH'T NO.	REV.	
350A-WW-SLD-ST1-N-8573	01 / 01	0	

